Scott Viteri

July 31, 2026

Phone: (480) 266-0242
Email: sviteri@stanford.edu
Google Scholar: Scott Viteri - Google Scholar

Profile

I am interested in training AIs to be human-like both in terms of the concepts they contain, as well as the values they hold. I think the key to this lies in training the language model to produce its own memory during training and to influence its future training data. In this vein, I have been optimizing pre-trained models to produce Chain-of-Thought reasoning which is *informative* to itself and to humans. In the future, I am interested in using a similar blend of unsupervised learning and reinforcement learning towards active learning, in the hopes that this sets the preconditions to raise the AI like a child, and have that result process in something child-like.

During my PhD, my focus has evolved from formal verification and programming languages to AI alignment. Prior to this, I majored in computer science and electrical engineering at MIT, contributing to AI and robotics research. After MIT, I explored interactive theorem proving at CMU with Simon Dedeo, publishing research on abduction in mathematics in the *Cognition* journal. My primary character trait is curiosity, and I really love math.

Education

Massachusetts Institute of Technology, BS Computer Science and Electrical Engineering **2014–2018**
Stanford University, Doctoral Candidate **2019–Current**

Employment

Stanford University, CS PhD Candidate, Stanford CA **9/2019–Current**

AI Safety / Alignment (1/2022 – Current)

- Developed Markovian Training (accepted to ICLR 2026), a method that routes reasoning through a bounded chain-of-thought bottleneck, improving GSM8K by 38 percentage points while producing more causally faithful explanations.
- Published Uncovering Latent Chain of Thought Vectors in Large Language Models at the ICLR 2025 Workshop on Weight Space Learning, showing that activation steering vectors can induce chain-of-thought reasoning without natural-language prompting.
- Created and twice taught Stanford course CS362: Research in AI Alignment.
- Won $10K in Alignment Research Center contest for Eliciting Latent Knowledge (ELK) solution proposals.
- Presented talks at the Topos Institute and the Alignable Structures conference on ontology maps.
- Mentored for AGI safety fundamentals reading group and TA'd for Stanford's AI Alignment courses (STS 10SI, STS 20SI).

Formal Verification / Programming Languages (9/2019 – 1/2022)

- Created a flexible method for shifting between higher-order and first-order logic in program verification, with Oded Padon, using the Veribetrkv key-value store.
- Published to IJCAR on proof production in CVC5 SMT solver.
- Co-wrote neural decoder auto-completion framework using context-sensitive predictions.

Carnegie Mellon University, PL Researcher, Pittsburgh PA **10/2018–8/2019**

- Extracted structural patterns from Coq proof trees to support automatic theorem proving research.
- Extended plugin to reify Coq AST, built a repository for analyzing ASTs from famous mathematical proofs.

- Published in *Cognition* (2022) on epistemic phase transitions in mathematical proofs, using a dataset of 48 machine-aided Coq proofs to study how deductive and abductive reasoning produce confidence in complex arguments.

MIT, Artificial Intelligence Researcher, Cambridge, MA **1/2018–6/2018**

- Worked on a Coq framework for knowledge representation and planning, treating sub-goaling as automatic theorem proving.
- Used coherent logic for theorem-proving "hints" in planning systems.

MIT, Robot Locomotion Group, Cambridge, MA **1/2017**

- Used Vicon camera systems to analyze performance of state estimators for Atlas and Valkyrie robots.

Amazon, Software Engineering Intern, Seattle, WA **6/2016–8/2016**

- Developed machine learning tools to predict user behavior on the Amazon website, reducing server requests with predictive modeling.
- Worked with Spark, Hadoop File Systems, and Elastic Map Reduce.

Roambotics, Software Development Intern, Phoenix, AZ **6/2015–8/2015**

- Designed and implemented architecture for the company's autonomous robot, integrating C and C++ libraries using Ant build tools.

Technical Skills

- **Computer Languages:** Python, Racket, Lean
- **Other Tools:** GNU/Linux (Ubuntu, Arch)
- **Hardware:** Comfortable with laser cutters, lathes, mills, 3D printers, and other shop tools.

Interests and Activities

Singing, guitar, Brazilian Jiu Jitsu, Zouk dancing, and math